Chapter 37
Crypto Derivatives

Chapter 36 worked through markets where the valuation machinery of Chapter 5 and Part III fails, and in each case it failed because of the underlying: electricity cannot be stored, an emissions allowance has a supply a legislature sets, a degree-day index cannot be held at all. This chapter's underlying breaks none of those rules. A bitcoin is storable at essentially no cost, perfectly fungible, divisible to eight decimals, pays no dividend, has no convenience yield, no seasonality, and no delivery constraint. By the standards of Chapter 36 it is the best-behaved underlying in the part—better behaved than crude oil, which has to be put somewhere, and far better behaved than an allowance whose validity is a political question.

So Chapter 5's carry relation applies without modification. With no income and no storage cost, the arbitrage-free futures price is $F = Se^{rT}$, the cash-and-carry arbitrage can be run in both directions, and the basis trade of Chapter 6 transfers across with the algebra unchanged. When market prices depart from that relation, as Section 37.1 shows they do by several percentage points a year, the reason is not the underlying.

What is missing is everything around the contract. There is no central counterparty, no segregation of client collateral, no settlement cycle that pauses trading (the offshore venues mark and liquidate continuously, with no pause in which to collect margin or manage a default), no enforceable netting opinion of the kind Chapter 27 called worth more than any risk model, and no lender of last resort. The pricing of Chapter 5 and Part III assumes an infrastructure it never names, and this is the market where that assumption is visible because it is absent.

The distinction between the contract and the infrastructure around it organizes the chapter, and it is not a technicality. The largest custody failure in the market's history was not caused by a mispricing, a model failure, or a move in the underlying, and Section 37.10 argues that describing it any other way misplaces the lesson.

Sections 37.1 to 37.5 take the contracts in turn: listed futures and their basis, the effect of a staking yield, perpetual futures, options, and the inverse contract with its choice of numeraire. Section 37.6 looks at the producer and Section 37.7 at the spot ETFs that narrowed the gap to the regulated market; Sections 37.8 and 37.9 are the decentralized venue and the settlement asset. Section 37.10 is the custody failure, and Section 37.11 says who trades these derivatives and why.

37.1 LISTED FUTURES AND THE BASIS

Regulated futures on bitcoin have traded since 2017, and the offshore venues list futures on hundreds of tokens. The pricing is Chapter 5's, unchanged.

Example 37.1 The basis trade, transplanted

Spot is \$60,000 and an offshore venue's three-month futures price \$61,800. Crypto markets quote the basis the other way round from Section 3.3's spot minus futures, and as a percentage: the futures premium over spot, $(F - S)/S$. Here that is $1{,}800/60{,}000 = 3.00\%$ over the quarter, or **12.55% annualized** with quarterly compounding $(1.03^4 - 1)$, against a dollar financing cost of 5%

a year on the same compounding basis. In Section 5.9's terms that 5% is the whole cost of carry, since a coin pays no income and costs nothing to store, while the futures contract is priced as if the cost of carry were 12.55%. Buying spot, selling the futures contract and waiting therefore earns the gap: **7.55%** a year over the cost of carrying the position.

Traders rarely put up the full position. Leverage here is the position's notional divided by the capital posted against it (for one unit of a futures contract, the futures price divided by the margin on it), so 20% initial margin is 5× leverage, and the return on that capital is roughly the 7.55% multiplied by the leverage (taking the coin itself as financed in full at the 5%):

Initial margin	Leverage	Return on capital
50%	2.0×	15.1%
35%	2.86×	21.6%
20%	5.0×	37.8%

This is Section 6.5's Treasury basis trade with the numbers moved up by two orders of magnitude, and the reason the excess over financing is 7.55% rather than a few basis points is not that the arbitrage is harder. It is that three things Chapter 6 took for granted are absent: there is no CCP, so the position is an unsecured exposure to a venue; margin is set at the venue's discretion rather than by a published risk model, and can be raised without notice; and the collateral is held by the counterparty, so the trade's return is conditional on the counterparty surviving the quarter.

The basis is therefore not an inefficiency but a price. This is Section 5.15's point that arbitrage needs a balance sheet, here one entrusted to the venue.

The return in Example 37.1 does not show the trade's main risk, which is a shortage of cash rather than a loss of value, as Example 37.2 shows.

Example 37.2 A hedged trade that a rally can close

The basis trade has no net exposure to the price: the spot leg and the short futures position offset, so the profit is locked in at entry. The cash flows are not flat. The futures position is marked to market as the price moves; the spot is not (unless the coin itself is posted as the margin, as an offshore venue allows). So a rally calls cash on one leg against an unrealized gain on the other.

At $60,000 spot the futures price is $61,800, and 20% initial margin on the contract is $12,360 per unit. The cash called is the variation margin the short futures position pays as the futures price rises; with the basis held at 3%, a rally raises the futures price by the same percentage of $61,800:

Rally	Cash called	Share of margin	Profit at expiration
+10%	$6,180	50%	unchanged
+20%	$12,360	100%	unchanged
+33%	$20,394	165%	unchanged
+50%	$30,900	250%	unchanged

A 20% rally, the margin rate itself, exhausts the margin on a position whose profit at expiration has not changed. The trade is liquidated for want of cash while the hedge is working exactly as intended, and the liquidation forfeits the 7.55%, which was locked in only for a position held to expiration. The futures loss is realized in cash at once, while the offsetting gain on the coin exists only as a mark until the coin is sold; if it is sold alongside, it recovers nearly all of that loss but none of the carry; if it is kept, it is an unhedged long.

This is Section 8.9's margin call as a liquidity risk, and the reader has met it twice already: it is Example 36.9's stack and roll, the problem that brought down Metallgesellschaft's oil hedge, and it is Chapter 8's central point, that a hedge can be correct in value and still fail on timing. What this market adds is that the venue sets the margin, so the schedule in the table is not a parameter of the trade but a decision someone else may revise mid-position.

Hence the leverage is not free. At 5× the headline return is 37.8%, and the rally that ends the

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trade is 20%. At 70% volatility, a stressed quarter for bitcoin (Example 37.7 below), a coin with no expected drift in its price rises 20% at some point within a quarter with a probability of about 55% (the touch probability of Section 34.3, at zero drift). Held this way, with the coin outside the margin account, the trade is more likely than not to be liquidated before the futures contract expires and the 7.55% is earned.

The direction of the basis is informative in the same way Chapter 36's curve was. A persistent premium says leveraged long demand exceeds the capital willing to finance it, and the premium collapses, occasionally inverting, when that demand unwinds. It shows how traders are positioned, measured by a market price rather than a survey, and it is easier to read than Chapter 36's convenience yield, which is never quoted and has to be backed out of the curve as a residual. Here the premium itself is quoted, and because a coin pays no income and costs nothing to store, subtracting the dollar interest rate leaves the excess over carry of Example 37.1 with nothing else to estimate.

37.2 WHEN THE UNDERLYING PAYS A YIELD

This chapter opened by saying the underlying pays nothing. That is true of bitcoin and false of ether, the token of the Ethereum network and the second-largest asset in the market, and the correction is worth making because it moves the carry relation from Chapter 5's case of an asset that pays nothing to its case of an asset with a known yield.

A **proof-of-stake** asset pays its holder for committing coin to validating the network. That is a continuous yield q on the asset itself, so the forward price is Chapter 5's known-yield case (Example 37.3 prices it):

$$F = Se^{(r-q)T}. \tag{37.1}$$

The yield q is not set by anyone's discretion. On Ethereum the protocol determines it, and it has two parts. The larger is newly issued coin paid to validators, at a rate that falls as more coin is staked: approximately $166.3/\sqrt{M}$ a year when M ether in total is staked, or about 2.5% at the roughly 44 million staked in late 2026. The more coin is committed, the thinner the reward is spread, so the yield regulates how much of the supply is staked. The smaller part is the transaction fees and the payments for ordering transactions that a validator collects when it proposes a block; these vary with network activity and typically add a few tenths of a percent. A staking provider keeps a commission, typically 10% to 25%, out of both.

Two consequences follow for (37.1). First, q is not known in advance, since both the amount staked and the network's activity move; a forward embeds the yield expected over its life, not a contractual rate. Second, the issuance part is paid in new coin, so it is partly a transfer from holders who do not stake, whose share of the supply shrinks. Unlike a dividend, it is not paid out of anyone's earnings, and a holder who does not stake is diluted, not merely forgoing income.

Example 37.3 A dividend the asset did not previously have

Spot \$3,000, a 5% dollar rate and the staking yield q, both continuously compounded, and three months, so the forward is $3{,}000\,e^{(0.05-q)\times 0.25}$:

Staking yield q	Forward	Quarterly basis	Annualized
0.0%	\$3,037.74	+1.258%	+5.13%
2.5%	\$3,018.81	+0.627%	+2.53%
3.5%	\$3,011.27	+0.376%	+1.51%
5.0%	\$3,000.00	0.000%	0.00%
7.0%	\$2,985.04	−0.499%	−1.98%

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The basis is computed as in Example 37.1: the premium $(F - S)/S$ over the quarter, annualized by compounding four quarters. At $q = r$ the futures price equals spot, and above it the curve inverts. The inversion says nothing about scarcity. The asset now pays a yield, and a persistent backwardation that would signal scarcity in Chapter 36 signals here only that the staking yield exceeds the dollar financing rate.

So Example 37.1's basis is not a clean positioning indicator for a staked asset. The observed premium mixes leveraged demand with the yield, and a desk reading it as sentiment will misread it whenever q moves. The two are separable only if the staking yield is quoted independently, which it is, so the correction is available to anyone who makes it.

The first US spot ether ETFs of Section 37.7 were launched without staking, and that has a cost. Grayscale's ether funds began staking in October 2025, and BlackRock's staked trust followed in March 2026, staking 70% to 95% of its ether and keeping the rest unstaked to meet redemptions. A fund holding coin without staking it forgoes the yield: on a $10 billion fund at about 2.5%, that is $250 million a year, and because most of the yield is new issuance, the fund's holders are diluted by it rather than merely missing it. In exchange it buys liquidity.

Staked coin is not immediately deliverable. Exiting a validating position means joining a queue (exit and withdrawal together usually take under a week, but took over 40 days in September 2025, when many validators exited at once), so the deliverable supply is smaller than the outstanding supply. A fund that must meet redemptions daily cannot hold all its assets behind any queue, which is why a staked fund keeps part of its coin unstaked, and the cash-and-carry arbitrage of Section 37.1 needs unstaked coin to close.

This qualifies the chapter's opening claim in a specific way. The underlying is storable at no cost *and* it now offers a yield for making it temporarily undeliverable, so a holder faces a genuine choice between earning q and retaining delivery capacity. That is Chapter 36's convenience yield reappearing—not as an inference from a curve, but as a posted rate a holder gives up to keep the asset liquid.

37.3 PERPETUAL FUTURES

A **perpetual futures** contract (a **perpetual** for short) has no maturity. It never delivers and never expires, so nothing forces convergence to spot. What replaces convergence is a payment: at fixed intervals, usually every eight hours (some venues hourly), longs pay shorts a **funding rate** set by the gap between the perpetual's price and a spot index, plus, on most venues, a small fixed interest component (0.01% per eight hours); when the perpetual trades far enough below the index the payment reverses.

The mechanism inverts the usual construction. A dated futures contract is held to spot by the *prospect* of delivery, and the cost of carry is embedded in the forward price. A perpetual is held to spot by an explicit cash flow instead, and that cash flow takes the place of the premium in a dated contract. A long in a dated contract pays the carry through the price: it buys at a premium over spot, and the premium shrinks to nothing by delivery. A perpetual's price never converges, so there is no premium to shrink, and without a payment a long perpetual would be leveraged exposure to the coin with nothing paid for the financing. The funding rate is that payment. It is not a cost of carry in Chapter 5's sense, a quantity fixed by interest rates and storage, but a price set by the demand for leverage. While the perpetual trades at a premium, the funding rate has to cover at least what an arbitrageur who sells the perpetual and holds the coin pays to hold it, and on top of that it carries whatever premium leveraged longs pay, just as the dated basis of Example 37.1 does. Example 37.4 converts one into the other.

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Example 37.4 Funding is the basis, quoted differently

Funding is paid every eight hours, so there are $3 \times 365 = 1{,}095$ periods a year:

Funding per 8 hours	Annualized (simple)	Equivalent 3-month premium
+0.0100%	+10.95%	+2.74%
+0.0300%	+32.85%	+8.21%
+0.0500%	+54.75%	+13.69%
−0.0200%	−21.90%	−5.48%

A quoted funding rate of 0.01% per eight hours and a three-month futures contract at a 2.74% premium are the same statement about the price of a levered long, and a desk running both should see them agree. When they do not, the difference is the cost of the two venues' credit and margin terms rather than anything about bitcoin. The first row is also the venues' default: inside a narrow band around the index the formula pays the fixed 0.01% whatever the gap, so funding reads demand only once it moves off that default.

Chapter 5 had to *infer* a convenience yield as the residual that made an equation hold, and Chapter 36 spent a section extracting one from a curve. The corresponding quantity here, the price of holding a levered long, is printed every eight hours rather than backed out of a curve.

The perpetual exists because a contract that never settles asks less of the infrastructure than a dated one: there is no expiration to settle and nothing for its holders to roll. The instrument is an adaptation to missing infrastructure, which is the chapter's theme in a single product. The adaptation has since come onshore: on May 29, 2026 the CFTC approved a cleared bitcoin perpetual for a US exchange. Inside a CCP the contract keeps its funding rate; what changes is everything around it.

Liquidation replaces the margin call. There is no cure period, and no one at a broker deciding whether to give the client more time. When margin falls below a threshold the venue closes the position at once. In traditional markets the same trigger usually produces a margin call instead (from the broker on exchange-traded futures, from the counterparty under a credit support annex on OTC derivatives, from the broker on a securities margin account), and a margin call starts a clock: the client has about a day to raise cash, and can see whether the move reverses before committing it. The difference lies with the venue, not the coin, since bitcoin futures on CME carry margin calls like any other futures contract, and have kept them since CME moved its crypto futures to round-the-clock trading on May 29, 2026. On an offshore venue the trigger ends the position. The client can add collateral at any moment up to that point, but must do so before knowing whether it will be needed, so protection means collateral posted early and left idle, or a position watched around the clock—and a fast market can cross the threshold before anyone can act. Section 8.9's collateral loop runs here in minutes rather than days. And because leveraged positions use a few standard leverage multiples, their liquidation levels cluster and the forced flow is a known function of the price: Section 35.7's barrier clustering in another market, with one difference that Example 37.5 draws out.

Example 37.5 A cascade with published barriers

Take \$12 billion of leveraged long perpetual positions: \$10.8 billion in the four tiers below and \$1.2 billion at lower leverage. A position at ℓ times leverage has margin worth $1/\ell$ of its notional, so it liquidates on a fall of roughly $1/\ell$ (a little less in practice, since the venue closes it before the margin is entirely gone):

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Leverage	Liquidates on	Notional in tier	Cumulative forced selling
25×	−4%	$1.2bn	$1.2bn
20×	−5%	$1.8bn	$3.0bn
10×	−10%	$4.2bn	$7.2bn
5×	−20%	$3.6bn	$10.8bn

The table assumes that every position's leverage is measured from the current price and that no one adds margin as the price falls. A 10% decline mechanically forces $7.2 billion of selling, which is 60% of those positions, and the selling arrives into a market already falling. The second-round effect is the point: that flow pushes the price further, which triggers the next tier. It is the sale at Section 35.6's knock-in barrier repeated at every tier, without the buying that precedes it there, and no issuer decides when to trade: the venue's automated liquidation system does.

But the disclosure is the opposite of Section 35.7's. Autocallable barriers cluster because issuers independently choose round fractions, and no issuer can see the aggregate; that was the whole difficulty. Here open interest and the venues' leverage schedules are public in real time, and data firms publish the liquidation profile estimated from them, so anyone can see its shape.

That does not make the cascade smaller. It changes who gains from the cascade: the flow is visible in advance to the traders best placed to trade ahead of it, and the resulting anticipation is itself part of why the moves are abrupt. Transparency relocated the problem rather than removing it, which is worth setting beside Chapter 2's case for uniform, transparent margin.

The largest recorded case is October 10, 2025, when, amid new US tariff announcements aimed at China, trackers such as CoinGlass counted about $19 billion of liquidations in roughly a day, and at least one large venue closed profitable positions by auto-deleveraging (Section 37.10).

37.4 OPTIONS

Options trade on the offshore venues, on CME futures since 2020 and, since 2024, on the regulated ETFs of Section 37.7. The pricing is Section 16.5's. The offshore contracts are European and cash-settled, and each is priced against the futures contract of the same expiration, so the model is Black's, with the futures price in place of spot; market makers hedge with that contract and with the perpetual. That choice matters more here than anywhere in Part III. Black's formula needs only the forward, so it takes in the basis of Section 37.1 and the staking yield of Section 37.2 without anyone having to estimate them, which is Section 16.8's rule of getting the forward right. A value computed from spot and the dollar rate assumes a forward of Se^{rT}, missing the part of the basis that financing does not explain, and so misprices the option, as Example 37.6 shows. Time is counted in calendar days over 365, since the market never closes.

Example 37.6 Pricing off the forward the market trades

Take spot at $60,000, the three-month futures price of $61,800 from Example 37.1, a strike of $60,000, 70% volatility and a 5% dollar rate, continuously compounded:

Forward used	Call	Put
Futures price, $61,800 (Black)	$9,273	$7,496
Se^{rT}, $60,755	$8,661	$7,916

Pricing from spot and the dollar rate undervalues the call by $612, or 7%, and overvalues the put by $420, or 6%. The two errors are the same mistake: put–call parity, $c - p = e^{-rT}(F - K)$, ties both prices to the forward, and the second row uses the wrong one. The hedge is in the same

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contract. The call's delta against the futures contract is $e^{-rT}N(d_1) = 0.595$, with

$$d_1 = \frac{\ln(61{,}800/60{,}000) + (0.70^2/2) \times 0.25}{0.70\sqrt{0.25}} = 0.2595,$$

so a dealer who sells the call buys 0.595 futures contracts per coin.

These options differ from an equity index's in three ways. The first is the level and the term structure. At-the-money implied volatility on bitcoin mostly ran at 35% to 55% in 2026, spiking above 80% in February (ether and smaller tokens higher), and short-dated volatility jumps above long-dated in a sharp rally as well as a sharp fall, where equity indices invert mainly in a sell-off (Section 19.6).

The skew changes sign. Equity index skew has pointed one way since 1987; crypto skew reverses with the market, with calls in demand in a rally and puts in demand in a decline. This is the pattern Section 19.1 found in currencies, whose tilt also changes sign as sentiment moves, rather than the one it found in equity indices, and for a similar structural reason: there is no leverage effect (the tendency of a stock's volatility to rise as its price falls, because a falling share price raises the firm's debt relative to its equity) and no large, persistent buyer of protection of the kind equity portfolios supply, so the skew reflects whichever side is currently buying options.

The volatility risk premium is large and unstable. Section 19.10's two to four volatility points on equity indices is often ten points or more here, which is why so much of the retail product in this market is a covered call in some form. What that level of volatility costs is worth seeing in currency rather than in percentage points.

Example 37.7 What a 70% volatility costs

Consider a thirty-day at-the-money straddle on a \$60,000 spot:

Volatility	Straddle	Share of spot	Break-even move
15%	\$2,059	3.43%	±3.43%
30%	\$4,116	6.86%	±6.86%
50%	\$6,857	11.43%	±11.43%
70%	\$9,591	**15.99%**	±15.99%
90%	\$12,318	20.53%	±20.53%

An equity index at 15% costs 3.43% of spot to straddle for a month; this market at a stressed 70% costs 15.99%. The buyer needs a 16% move in thirty days merely to return the premium, and the seller collects 16% of spot for a month of risk, which is why the yield products of Section 37.11 can advertise returns that look impossible and are not.

The premium is real and it is not a riskless profit. To first order a delta-hedged short option earns, per year,

$$\frac{1}{2}\,\Gamma S^2 \left(\sigma^2_{\text{implied}} - \sigma^2_{\text{realized}}\right),$$

so per unit of dollar gamma ΓS^2 it earns $\frac{1}{2}(\sigma^2_{\text{implied}} - \sigma^2_{\text{realized}})$, which for the first row of the table below is $\frac{1}{2}(0.70^2 - 0.60^2) = 6.5\%$:

Implied	Realized	Profit per unit of ΓS^2
70%	60%	+6.5% a year
70%	55%	+9.4% a year
70%	75%	−3.6% a year

Five points of adverse surprise cost 3.6% a year, so the strategy's sign depends on a quantity

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no one observes in advance, which is Section 22.1's point about delta-hedged options, with larger numbers.

37.5 INVERSE CONTRACTS AND THE CHOICE OF NUMERAIRE

A large part of this market trades in contracts that settle in the coin rather than in dollars, and the convention is not cosmetic. An **inverse** contract has a notional fixed in dollars, margin posted in the coin, and settlement paid in the coin: a long position of notional Q entered at F_0 pays

$$\Pi_{\text{coin}} = Q\left(\frac{1}{F_0} - \frac{1}{S_T}\right). \tag{37.2}$$

This is Section 28.6's numeraire question posed by a contract specification rather than by a model. Convert (37.2) at the terminal price and the dollar payoff is $Q(S_T/F_0 - 1)$—*exactly linear*. Leave it in coin and it is not, as Example 37.8 shows.

Example 37.8 Linear in one currency, curved in the other

Take a long inverse position, \$100,000 notional, entered at \$60,000:

Price	Move	P&L in coin	P&L in dollars
\$30,000	−50%	−1.6667	−\$50,000
\$42,000	−30%	−0.7143	−\$30,000
\$48,000	−20%	−0.4167	−\$20,000
\$72,000	+20%	+0.2778	+\$20,000
\$90,000	+50%	+0.5556	+\$50,000
\$120,000	+100%	+0.8333	+\$100,000

The dollar column is a straight line and the coin column is not. A 50% fall costs **1.6667 coin** while a 50% rise earns only 0.5556, and the coin gain is capped at $Q/F_0 = 1.6667$ however far the price rises while the coin loss is unbounded. A long inverse position is short gamma measured in coin, with $\partial^2\Pi_{\text{coin}}/\partial S^2 = -2Q/S^3$.

With an initial margin of 0.5 coin, the long's equity is $0.5 + Q(1/F_0 - 1/S)$ coin, which moves as follows:

Price	Equity (coin)	Equity (dollars)
\$54,000	+0.3148	\$17,000
\$48,000	+0.0833	\$4,000
\$42,000	−0.2143	−\$9,000

At entry the 0.5 coin is \$30,000, so a dollar-margined trader at this leverage is wiped out by a 30% fall; in coin the equity is gone where $0.5+Q(1/F_0-1/S) = 0$, at $S = 1/(1/60{,}000+0.5/100{,}000) =$ \$46,154, a 23% fall. Equity falls faster than a dollar-margined trader's (\$17,000 against \$20,000 at \$54,000), because the margin is itself coin and loses value with the price, so liquidation arrives earlier than a dollar-margined trader would compute from the same leverage. The collateral and the exposure are the same asset, which is Chapter 27's wrong-way risk in its purest form, written into the contract rather than arising from a correlation.

The convention has a practical origin. An inverse contract needs no dollar leg, so a venue can run it without touching the banking system: the whole contract lives in the asset it references. That mattered when venues had no bank accounts, and it is why the oldest contracts in this market are inverse ones; the newer contracts margined in dollars or in dollar tokens (stablecoins, Section 37.9) now carry most of the volume. It also suits a miner, which can sell one against its coin and fix the coin's dollar value without ever holding a dollar.

The pricing follows Chapter 28. Let $F = Se^{rT}$ be the carry forward of Chapter 5, the

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frictionless market's price, setting aside the venue credit that Section 37.1 found in the basis; because the discounted coin price is a martingale under the dollar risk-neutral measure, F is also $E[S_T]$, the expectation under that measure. In dollars the payoff $Q(S_T/F_0 - 1)$ is linear and worth

$$e^{-rT} Q\left(\frac{E[S_T]}{F_0} - 1\right) = e^{-rT} Q\left(\frac{F}{F_0} - 1\right),$$

which is zero at $F_0 = F$, so the fair entry price is the ordinary forward. With the coin as numeraire, a value in coin is the coin-measure expectation of the payoff in coin, and the change of measure has density $(S_T/S)\, e^{-rT} = S_T/F$, so

$$E^S\left[\frac{1}{S_T}\right] = E\left[\frac{S_T}{F} \times \frac{1}{S_T}\right] = \frac{1}{F},$$

and the contract is worth

$$Q\left(\frac{1}{F_0} - \frac{1}{F}\right) \text{ coin},$$

zero at the same price. Mixing the two measures gives a wrong answer. For a lognormal price with volatility σ, $\ln S_T$ has mean $\ln F - \sigma^2 T/2$ and variance $\sigma^2 T$ under the dollar measure, so averaging the coin payoff's $1/S_T$ with dollar-measure probabilities gives

$$E\left[\frac{1}{S_T}\right] = e^{-\ln F + \sigma^2 T/2 + \sigma^2 T/2} = \frac{e^{\sigma^2 T}}{F},$$

not $1/F$. The extra factor $e^{\sigma^2 T}$ is Section 30.6's quanto term, with $1/S_T$ as the asset and S_T as the exchange rate, so their correlation is -1, and no correct valuation of this contract contains it.

37.6 THE PRODUCER: MINING AS A SPREAD POSITION

This market has a producer, as Chapter 36's power market does. A miner spends electricity and capital to receive coins, so it is long the coin and short power: Section 36.6's spark spread with a different output. The hedging problem lies where the comparison fails (Example 37.9).

Example 37.9 A miner's break-even, and the hedge the protocol supplies

A miner produces 500 coins a year against \$14 million of electricity and \$4 million of other costs. Break-even is \$36,000 a coin, and because the costs are fixed while revenue moves with the price, the profit moves much more than the price does:

Coin price	Revenue	Costs	Profit	Profit/revenue
\$30,000	\$15.0m	\$18m	−\$3.0m	−20.0%
\$45,000	\$22.5m	\$18m	+\$4.5m	+20.0%
\$60,000	\$30.0m	\$18m	+\$12.0m	+40.0%
\$90,000	\$45.0m	\$18m	+\$27.0m	+60.0%

The output is not a fixed quantity, and that is what separates this from the spark spread. The network issues new coins at a fixed rate, a batch with each block of transactions, and a miner wins blocks in proportion to its share of the **hash rate**, the computing power all miners together apply to the puzzle each block requires. The protocol sets how hard that puzzle is, and its **difficulty adjustment** resets it at intervals so that blocks keep arriving on schedule however much hardware is running. If a price fall drives competitors off, the survivor's share rises at once but blocks slow down, so its output rises only when the difficulty falls and blocks return to schedule.

Suppose the price falls from \$60,000 to \$42,000 and 20% of the network's hash rate leaves:

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	Revenue	Change
Before	\$30.0m	—
Before the difficulty adjustment	\$21.0m	−30%
After the difficulty adjustment	\$26.3m	**−12.5**%

The protocol converts a 30% revenue shock into a 12.5% one. The miner is long the price, and it is also long the exit of competitors, which happens when the price falls; the two offset, so the exposure is to price × share, and share moves against price.

Three qualifications keep this from being a free hedge:

- the difficulty adjustment comes at intervals of roughly two weeks, so the offset arrives after the cash-flow problem it would have solved;
- the protection comes from the miners who leave, and those are the high-cost ones, so a high-cost miner is near the exit itself when it needs protection most, while the low-cost miner, which needs it least, gains the most; and
- the hedge works through quantity, so a forward sale of a fixed amount fits neither direction: after a fall the miner produces more than it sold and the surplus is unhedged, and after a rise it produces less and must buy coin at the high price to deliver.

This explains how miners that hedge do so. They sell a fraction of expected output forward rather than all of it, and favor puts over forwards, because a forward that must be delivered into is a short position in a market where the quantity is itself uncertain. This is volume risk. The contract built for it is the swing contract of Section 36.5, in which the buyer chooses how much to take within limits; a miner would want the mirror image, with the choice of how much to deliver on its own side. The nearest listed contract is a hashprice futures contract, which fixes revenue per unit of hash rate rather than per coin, and it trades thinly, so most miners approximate the hedge with a strip of listed puts and accept the mismatch.

Power contracting is the other half of the position and it is where the interesting structures are. A miner with an interruptible power contract has sold an option to the grid: it agrees to shut down on demand in exchange for a lower price. Interruptible contracts are old (aluminum smelters have signed them for decades), but a miner is an unusually good counterparty for one, because what it gives up by shutting down is coin: the shutdown cost is observable at every instant as the coin price times the forgone production, less the power not bought, so the option's exercise value can be read from a quoted price. Several grids now contract with miners this way.

37.7 SPOT ETFS AND THE NARROWING OF THE GAP

The most consequential development in this market was not a derivative. United States spot bitcoin exchange-traded funds began trading in January 2024, spot ether funds in July 2024, and options on the bitcoin funds in November 2024.

What that changed is precisely the thing this chapter says is missing. An ETF share sits in ordinary custody, its options clear through the Options Clearing Corporation, the positions net under an enforceable framework, and the collateral is segregated. None of it altered bitcoin itself; it altered the infrastructure wrapped around it.

The gap between the two venues is therefore measurable, and it is the price of that infrastructure. An option on the ETF and an option on the offshore venue reference nearly the same underlying and differ in counterparty, segregation, netting, and margin treatment. Net of the contract differences (the ETF options are American, settle in fund shares and trade only in US hours) and of any difference in liquidity, the spread between their implied volatilities is Chapter 27's capital and Chapter 8's collateral, expressed in volatility points, and it is the cleanest available estimate of what the missing infrastructure was worth.

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37.8 AUTOMATED MARKET MAKERS

An automated market maker, the commonest design of decentralized exchange, has no order book. It holds an inventory of two assets and quotes by a rule: the **constant-product** rule keeps

$$x\,y = L^2,$$

where x is the inventory of a token, y the inventory of a dollar stablecoin, counted here as dollars, and L is set by what liquidity providers have deposited: a trade moves x and y along the curve and, apart from the fee it leaves in the pool, leaves their product, and so L, unchanged. $L = \sqrt{xy}$, the geometric mean of the two inventories, measures the pool's size and is called its **liquidity**; a deeper pool moves its price less for a trade of a given size. A stablecoin is a token its issuer promises to redeem for one dollar, backed by a reserve of dollar assets, and is the settlement asset of Section 37.9. Anyone may trade against the pool at the price the rule implies. That price is the ratio of the inventories, $S = y/x$, because along the curve $xy = L^2$ a small trade exchanges dx tokens for $dy = -(y/x)\,dx$ dollars. A buyer of the token pays stablecoin in and takes tokens out, so y rises, x falls and the price y/x goes up.

Providers of that inventory are told they are earning fees. They are, and Example 37.10 shows what they are paying for them.

Example 37.10 Providing liquidity is selling gamma

At a price S the two conditions $xy = L^2$ and $y/x = S$ pin the inventory to $x = L/\sqrt{S}$ and $y = L\sqrt{S}$, so the position is worth

$$V(S) \;=\; x\,S + y \;=\; 2L\sqrt{S}. \tag{37.3}$$

Two derivatives tell the whole story. The delta is $V'(S) = L/\sqrt{S} = x$, exactly the token inventory, so the pool's exposure to the price is always its current holding, and the rule rebalances that holding for it: it sells the token as the price rises and buys as it falls. And

$$V''(S) \;=\; -\frac{L}{2\,S^{3/2}} \;<\; 0,$$

so the position is concave in the price everywhere. A liquidity provider is short gamma, and Chapter 18 established what a short gamma position earns and what it loses.

Against simply holding the initial basket the pool falls short, and the market calls the shortfall **impermanent loss** (IL), a name suggesting the loss will reverse, which it does only if the price returns to where it started. The shortfall depends only on the ratio $k = S/S_0$ of the current price to the price S_0 at deposit. The inventory deposited, $x_0 = L/\sqrt{S_0}$ and $y_0 = L\sqrt{S_0}$, would be worth $x_0 S + y_0 = L\sqrt{S_0}\,(1+k)$ if simply held, while by (37.3) the pool is worth $2L\sqrt{S} = 2L\sqrt{S_0}\,\sqrt{k}$. Impermanent loss is measured as the pool's value relative to the held basket's, $\mathrm{IL} = V(S)/(x_0 S + y_0) - 1$:

$$\mathrm{IL}(k) \;=\; \frac{2L\sqrt{S_0}\,\sqrt{k}}{L\sqrt{S_0}\,(1+k)} - 1 \;=\; \frac{2\sqrt{k}}{1+k} - 1$$

Price ratio	0.25×	0.5×	0.8×	1.25×	2×	4×
Shortfall	−20.00%	−5.72%	−0.62%	−0.62%	−5.72%	−20.00%

It is symmetric in k and $1/k$: the shortfall depends on the size of the move in log terms, not on its direction, because neither token is privileged. And a payoff concave in the underlying is a short position in options, here a written straddle.

The cost of being short gamma, the expected rate at which the pool falls behind the basket when the price has no drift, which the market calls its bleed, has a simple closed form. Chapter 18's gamma cost per unit time is

$$\frac{1}{2}\,|V''|\,\sigma^2 S^2 = \frac{1}{2}\,\frac{L}{2S^{3/2}}\,\sigma^2 S^2 = \frac{L\sigma^2\sqrt{S}}{4},$$

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and dividing by the position's value $2L\sqrt{S}$:

$$\text{cost per year} = \frac{\sigma^2}{8} \quad \text{of the position, whatever the price and whatever the pool size.} \tag{37.4}$$

Volatility	40%	60%	80%	100%	120%
Fee yield needed to break even	2.00%	4.50%	**8.00%**	12.50%	18.00%

The expected value of the shortfall also has an exact closed form for a driftless lognormal price, $e^{-\sigma^2 T/8} - 1$, because the bleed is charged on the pool's current value and so compounds: at 80% volatility the exact loss over three months is 1.980% against $\sigma^2 T/8 = 2.000\%$, and over a year 7.688% against 8.000%. Multiplying the rate by the horizon gives only the first term of the exponential's expansion, so it always overstates the loss: by 4% of the exact figure at one year, but at 120% volatility over three years, where $\sigma^2 T = 4.32$, its 54.0% overstates the exact 41.7% by 29%.

So the question a liquidity provider should ask is the one an option seller asks: is the fee income above or below the cost of the volatility I am short? At 80% volatility the pool must return 8% a year in fees before the provider breaks even, and a pool advertising a 6% fee yield on a pair that volatile is advertising a loss.



Figure 37.1: Left: pool value against holding the same initial basket, normalized to 1. Right: (37.4).

Figure 37.1 is the reason this section is in a derivatives textbook rather than a technology one. The left panel is a written straddle drawn in unfamiliar coordinates, and the concavity is the product, not a defect of it. A provider who understood the position as a short option would ask about the premium; one who understands it as "earning yield" will not.

The concentrated-liquidity variant makes the option clearer. Providers may confine their inventory to a price range, which concentrates the gamma and raises both the fee income and the bleed—and outside the range the position is entirely one asset and the delta is constant. That is a strip of options written at strikes inside the range, with fees that accrue only while the price stays in it, like Section 33.6's range accrual; it is sold to retail users with no options vocabulary at all.

37.9 THE SETTLEMENT ASSET

Most trades on the offshore venues settle in a **stablecoin** rather than in dollars, and that changes what the quoted prices mean. A perpetual quoted at 60,000 is quoted in a token that its issuer promises to redeem at a dollar, so the futures price carries the issuer's credit whether or not anyone prices it that way.

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The peg is not a law of the market; it is an arbitrage with a cost. If the token trades below a dollar, someone buys it and redeems it at par for the profit. That trade works only while redemption works, which requires the reserve to be liquid, the issuer to be operating, and the redemption window to be open. Remove any of the three and the token stops being a dollar and becomes a claim on a portfolio, valued as Part V values any other claim.

Example 37.11 What the discount was pricing

In March 2023 Circle disclosed that $3.3 billion of the roughly $40 billion reserve behind its USDC stablecoin, or 8.25%, sat at a bank that had just failed. Take the rest of the reserve as sound and price the token as a claim:

Recovery on the exposed reserve	Fair value	Discount
100%	1.0000	0.00%
50%	0.9588	4.13%
0%	0.9175	8.25%

The token traded to about 0.8774, a 12.3% discount, against an arithmetic worst case of 8.25%. No recovery assumption produces that price, so the extra 4.0 points were not pricing the disclosed exposure at all.

They were pricing the redemption mechanism itself. The arbitrage that holds the peg needs an open window; a holder who fears the window closes before their turn will sell at any price above what they expect a queue to yield. So the discount decomposes into a credit loss on a known exposure and an option-like charge for access to the redemption facility, and only the first is in the disclosure.

The link to the rest of the book is direct. The failure of Silicon Valley Bank (Section 40.14) reached the derivatives market of this chapter through the settlement asset—not through any position, model, or counterparty in it. A trader perfectly hedged in coin terms lost 12.3% on the units the hedge was denominated in, which is Section 7.8's currency basis argument with the funding currency issued by a private firm.

Two consequences follow for anyone marking a book in this market. First, a stablecoin-denominated profit and loss is not a dollar profit and loss, and the two diverge exactly when the market is stressed—which is when the distinction matters. Second, margin posted in a stablecoin is collateral with credit risk and correlation to the very events that trigger margin calls, so Chapter 27's wrong-way risk applies to the collateral rather than to the exposure. A haircut on the settlement asset is the obvious remedy, and where venues apply one it is a percent or two, because a larger one would concede that the unit of account is not a dollar. Regulation is now supplying part of what the haircut did not. The US GENIUS Act of 2025, whose rules take effect on January 18, 2027, requires one-for-one reserves in cash, deposits and short-dated Treasuries, and redemption at par, and CFTC staff letters since December 2025 let futures commission merchants accept payment stablecoins, including those of national trust banks, as customer margin.

37.10 THE VENUE IS THE COUNTERPARTY

The failure of FTX in November 2022 is usually told as a story about crypto. It is more useful as a story about Chapter 2, because every layer that chapter described was missing and the absences can be listed one by one.

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Case Study 37.1 A default waterfall with no layers

Chapter 2 set out the sequence in which a clearing house absorbs a member default. Set that sequence beside what a large crypto venue had in place of each layer:

Chapter 2's waterfall	The venue
Defaulter's initial margin	Held by the venue, unsegregated
Defaulter's default fund contribution	None
The CCP's own capital, ahead of members	No distinction between the venue and its operator
Surviving members' mutualized fund	An "insurance fund" sized and used at the venue's discretion
Assessments and recovery tools	Losses socialized by unilateral decision

Bitcoin settled every block throughout. No contract mispriced, no model failed, and the underlying did what it always does. Client assets were not segregated, so a customer's exposure was not only to the market but to the venue, and the venue was under the same control as the trading firm on the other side of its customers' trades.

Section 27.2 called a jurisdiction's netting opinion (a lawyer's opinion that close-out netting will be enforced if a counterparty fails) worth more to a dealer than any risk model. This market shows what that means by its absence. There was no netting opinion, no bankruptcy-remote structure, and no supervisor that required either, so the entire apparatus of Part V (exposure at default, close-out netting, the margin period of risk) had nothing to attach to.

The lesson is not that the instruments were exotic. They were plain instruments: futures, perpetuals, and spot balances. What was exotic was the absence of the infrastructure every earlier chapter assumed.

Once the liquidated trader's margin and the venue's insurance fund are gone, a crypto venue uses **auto-deleveraging**, whose counterpart Part I's clearing houses keep as a last resort. When a liquidation cannot be filled at a price that leaves the venue whole, the shortfall is met by closing profitable positions on the other side, ranked by their profit and leverage. A trader who is right about the market and fully margined can have the position taken away to fund somebody else's loss, which is Chapter 2's tear-up brought forward to the layer after the insurance fund, without the mutualization, the disclosure, or the governance.

37.11 WHO TRADES CRYPTO DERIVATIVES, AND WHY

Leveraged directional traders dominate trading and are the perpetual's main users: a contract with no expiration, no delivery, and high leverage is the cheapest way to express a view. Their liquidation levels are the forced flow of Section 37.3, which makes their aggregate position a market-structure variable rather than a private one.

Basis traders and market-neutral funds harvest the basis of Example 37.1, and they are the capital that keeps the futures anchored to spot. They are also the first to leave, since their return is a spread and their risk is the venue, so the basis widens when the venue is in doubt.

Miners are the natural commercial hedgers, with a revenue stream in coin and costs in ordinary currency, chiefly for electricity, which is the position of Chapter 36's generator. Those that hedge the coin revenue do so by selling forward and buying puts.

Companies and funds holding coin sell covered calls for yield, giving up the price rise above the strike for a premium. It is the same trade, with the same asymmetry, as the yield products of Chapter 35.

Liquidity providers supply the inventory of Section 37.8, mostly without pricing the option

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they have written.

The market's participant mix is unusual in one respect. In most markets this book describes, the natural hedger came first and the speculator followed. Here the speculative flow built the market and the commercial hedgers arrived afterward, which is why the instruments are optimized for leverage rather than for delivery, and why the infrastructure lagged the volume by a decade.

SUMMARY

The underlying is the easy part and the infrastructure is the hard part. A bitcoin is storable at no cost, fungible, divisible, and pays nothing, so Chapter 5's $F = Se^{rT}$ is the arbitrage-free price. What is absent is the CCP, the segregation, the settlement pause, the netting opinion, and the lender of last resort: the infrastructure Chapter 5 and Part III assume without naming it.

That explains the prices. A futures basis far wider than the cost of financing is the venue's credit being priced, not an inefficiency. On a staked asset the basis also mixes in the yield, so it no longer reads as positioning alone. Options are priced off the forward the market trades rather than off spot. The perpetual is the only genuinely new contract: with nothing forcing convergence, a funding rate paid every eight hours does delivery's work and prints a price that Chapter 5's markets leave to be inferred. Liquidation replaces the margin call, and because open interest and leverage schedules are public, the forced flow can be estimated in advance, which relocates Section 35.7's barrier clustering rather than removing it.

Two conventions carry more risk than they appear to. An inverse contract is linear in dollars and curved in the coin, and its margin is coin, so liquidation arrives earlier than dollar leverage suggests. And most offshore trades settle in a stablecoin, so the quoted price carries an issuer's credit and the value of access to redemption. In both cases collateral and exposure are the same risk: Chapter 27's wrong-way risk sitting in the contract and in the unit of account.

The producer and the liquidity provider carry options they do not price as such. A miner holds Section 36.6's spark spread, long the coin and short power; the difficulty adjustment is an automatic hedge that arrives late and works through quantity, which is why miners sell only part of their output forward and lean on puts, and a miner on an interruptible power contract has sold the grid an option to shut it down. Providing liquidity to a constant-product pool is a written straddle that bleeds $\sigma^2/8$ a year whatever the price, so only substantial fees let the provider break even at this market's volatilities.

FTX is a story about Chapter 2. Every waterfall layer was missing, yet bitcoin settled every block throughout: no contract mispriced and no model failed. Client assets were not segregated, so the exposure was to the venue as well as to the market. The regulated route of the 2024 ETFs supplies the missing infrastructure. The instruments were never the problem.

FURTHER READING

The primary sources in this market divide sharply: venue documentation of funding-rate and liquidation mechanics shows how the contracts actually behave and is worth reading directly; most of the valuation literature applies Chapters 5 and 15 to the new contracts. On the perpetual contract, He, Manela, Ross and von Wachter and Ackerer, Hugonnier and Jermann derive the price the funding rule implies; they and the venues' specification documents are the place to start, before the secondary literature. On automated market makers, Milionis, Moallemi, Roughgarden and Zhang on loss-versus-rebalancing derive the $\sigma^2/8$ rate of Section 37.8 from the Black–Scholes setting, and the connection to Chapter 18's gamma cost is immediate. The FTX bankruptcy filings are the account of what Section 37.10 describes, and they enumerate what was and was not held. On the regulated products, the fund prospectuses set out the custody and creation mechanics that Section 37.7 argues are the whole of the change.

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PRACTICE QUESTIONS

37.1 Using Example 37.1, state the annualized basis and the return on capital at 20% initial margin, and give three reasons the spread is wider than Chapter 6's Treasury basis.

37.2 Using Example 37.2, state the rally that exhausts the posted margin on a fully hedged basis trade, and explain why a position that has lost nothing can be closed.

37.3 Explain why the leverage in Example 37.1 and the rally in Example 37.2 must be quoted together, and which earlier chapter's case has the same shape.

37.4 Using Example 37.3, state the forward price and basis at a 3.5% staking yield, and the yield at which the futures price equals spot.

37.5 Explain why a persistent backwardation in a staked asset does not mean what the same shape means in Chapter 36, and what must be quoted to separate the two.

37.6 Explain what holds a perpetual futures contract to spot in the absence of delivery, and why the mechanism is an adaptation to missing infrastructure.

37.7 Using Example 37.4, convert a funding rate of 0.03% per eight hours into an annualized rate and into the equivalent three-month futures premium.

37.8 Explain why liquidation on a crypto venue is not a margin call, and what the client gives up when it replaces one.

37.9 Using Example 37.5, state the forced selling triggered by a 10% decline, and explain why publishing the liquidation profile does not make the cascade smaller.

37.10 Explain why crypto skew changes sign with the market while equity index skew does not, and name the chapter whose pattern it resembles.

37.11 Using Example 37.8, state the coin profit and loss of a long inverse position at a 50% fall and at a 50% rise, and explain why the two are not symmetric.

37.12 Explain why an inverse contract's valuation is a change of numeraire, and what convexity term a desk introduces by averaging the coin payoff under the dollar measure.

37.13 Using Example 37.9, state the break-even coin price, and the revenue change from a 30% price fall before and after the difficulty adjustment.

37.14 Explain why the difficulty adjustment is only a partial hedge, giving the three qualifications, and why a hedging miner prefers puts to forwards.

37.15 Using Example 37.11, compute the fair value at 50% recovery on the exposed reserve, and explain what the observed 12.3% discount priced that the disclosure did not.

37.16 Explain why margin posted in a stablecoin is wrong-way risk in the collateral, and what remedy Section 37.9 says venues apply, where they apply it at all, at only a percent or two.

37.17 Using Example 37.10, state the shortfall at a price ratio of 2 and of 4, and explain why the function is symmetric in k and $1/k$.

37.18 Derive the $\sigma^2/8$ bleed of (37.4) from Chapter 18's gamma cost, and explain why it does not depend on the pool size.

37.19 Explain why the FTX failure is described as a custody failure rather than a market failure, and what the underlying did throughout.

37.20 Explain what auto-deleveraging is, how it differs from its analog in Part I, and why a correctly positioned and fully margined trader can still lose the position.

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FURTHER QUESTIONS

37.21 Example 37.4 treats the funding rate and the futures basis as two quotations of one price. Construct the arbitrage that enforces the relation, determine what frictions would let the two persist at different levels, and estimate how large a gap those frictions could sustain.

37.22 Example 37.5 treats the liquidation profile as static. Model it dynamically, so that the selling from one tier moves the price into the next, determine the condition on the leverage distribution under which the cascade is self-limiting rather than self-sustaining, and state what a venue could change to enforce that condition.

37.23 Example 37.9 shows that the difficulty adjustment offsets part of a price fall with a lag. Formulate the miner's revenue as a product of price and share, model the share as a delayed response to price, and determine the forward-sale fraction that minimizes the variance of revenue.

37.24 Example 37.11 decomposes a discount into a credit loss and a charge for redemption access. Build a model of the second component as an option on the window's remaining capacity, calibrate it to the observed 12.3%, and state what the calibrated parameter would have to mean.

37.25 Example 37.10 values a pool under geometric Brownian motion. Redo the analysis under a jump-diffusion, determine whether the $\sigma^2/8$ result survives, and state what a provider should charge for the jump component that the diffusion formula omits.

37.26 Concentrated liquidity confines the inventory to a price range. Formulate the resulting position as a strip of written options with the fee stream as a range accrual, derive the fee yield needed to break even as a function of the range width, and determine the width that maximizes expected profit at a given volatility.

37.27 Section 37.7 argues the ETF-to-offshore volatility spread prices market infrastructure. Design the measurement: state which contracts to compare, what must be controlled for, and how you would separate the infrastructure premium from a pure liquidity difference.

37.28 The snapshot in Section 37.10 maps a crypto venue onto Chapter 2's waterfall. Design a waterfall that would work for a venue with no members in Chapter 2's sense, state who funds each layer, and identify which layer is hardest to fund and why.

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